

Maritime Oil-Spill Attribution Intelligence

Complete External API Keys & Operational Credentials Specification

System Version: 2.4.0 (Production) | SIH26143 Forensic Core | Universal UTC Lineage

1. Master API Credentials Matrix

Variable Name	Service / Provider	Feature Unlocked	Status
AISSTREAM_API_KEY	aisstream.io	Real-time satellite/terrestrial live vessel streaming (/app/live)	CONFIGURED (Free)
CDSE_CLIENT_ID	Copernicus Data Space	Sentinel-1 SAR GRD satellite imagery search & ingestion	Optional (Free EU)
CDSE_CLIENT_SECRET	Copernicus Data Space	OAuth2 client authentication for CDSE satellite API	Optional (Free EU)
COPERNICUS_MARINE_USERNAME	Copernicus Marine (CMEMS)	Global Ocean Physics numerical hydrodynamic current grids	Optional (Free EU)
COPERNICUS_MARINE_PASSWORD	Copernicus Marine (CMEMS)	HTTP Basic Auth for CMEMS Subsetter / WMS current vectors	Optional (Free EU)
JWT_SECRET	Internal Platform	Cryptographic HS256 token signing for user sessions	Active (Secure Fallback)
CORS_ORIGINS	FastAPI Security	Whitelisted domains: maritime-iota.vercel.app, localhost	Active (Configured)

2. Live AIS Stream (AISSTREAM_API_KEY)

Operational Status: Configured on Render. The backend architecture implements a non-blocking asyncio.Queue that decouples frame reception from processing, preventing TCP buffer starvation and socket disconnects.

Provider: aisstream.io (Global Terrestrial & Satellite Marine AIS aggregation).

Origin Policy: Standard WSS protocol without artificial browser origin headers, eliminating Cloudflare bot-challenge rejections.

3. Copernicus Data Space Ecosystem (CDSE) Registration Guide

CDSE provides programmatic access to European Space Agency (ESA) Sentinel-1 C-band Synthetic Aperture Radar (SAR) imagery.

Step 1: Navigate to <https://dataspace.copernicus.eu> and click Register.

Step 2: Sign in and navigate to Identity Management:
<https://identity.dataspace.copernicus.eu/auth/realms/CDSE/account>

Step 3: Under **OAuth Clients**, click **Create Client** and obtain your **CDSE_CLIENT_ID** and **CDSE_CLIENT_SECRET**.

Step 4: Add both keys to your Render environment variables.

4. Copernicus Marine Service (CMEMS) Registration Guide

CMEMS provides multi-year and real-time ocean current velocity grids (u, v vectors at 0.083° resolution) driving the Lagrangian drift model.

Step 1: Visit <https://marine.copernicus.eu> and register a free EU account.

Step 2: Your registered login email/username and password serve directly as the API credentials.

Step 3: Add COPENICUS_MARINE_USERNAME and COPENICUS_MARINE_PASSWORD into Render.

5. Watermark-Free Dark Naval Cartography

The Leaflet investigation and live maps utilize **Esri World Dark Gray Canvas Base and Reference layers**. This cartography is 100% free, requires zero tokens or API keys, and completely eliminates map watermark artifacts.